DISCUSSION

IDENTIFICATION OF PMM2 AND THE DISEASE-ASSOCIATED VARIANT

The change of nucleotides C to T in the position 710 caused that the gene codified for another amino acid, the Methionine instead of Threonine with different properties. Even Though they are both neutral, Especifically, Threonine is polar and has an hydroxyl group(OH-) which allows it to form hydrogens bonds with Thr16. But , it is replaced by Methionine which can't form them and it's not polar. So, Methionine can't form a bond with Thr16 disrupting the stability which can affect the way the protein(enzyme) functions. Studies determined that changing a Threonine by a Methionine indeed has an effect on the stability of the protein, its interactions, and the enzymatic activity leading to malfunction. We can't assure that all the structure of the protein will be affected, but the interactions. So, it affects the function of PMM2 , which is to produce mannose-1-phosphate from mannose-6-phosphate important for GDP-mannose to take part in the glycosylation.
Also, we identified that there is a Mg2+ binding site inside the alpha- helix where our change of aminoacid is located. Mg2+ helps to conduct the enzymatic reaction of PMM2. The substitution could affect the molecular interactions . However, we don't have enough proof to demonstrate that the Mg 2+ union has been modified and regarding the results of RDSM the structure has been kept quite alike.

Moreover, with all the analysis, the consequences of the variant T237M are involved in a malfunction of the protein and glycosylation provoking an insufficiency for the organism leading to the disease PMM2-CDG.

STRUCTURAL COMPARISON OF WILD-TYPE AND T237M PMM2

From the superposition of 7O4G and 7O58 we obtained a low RMSD (0.352 Armstrong). This means that the positions of atoms in the main structure of the protein are very similar. The backbone is closely related to the RMSD, and consequently it's also low so the structure has little variations having a similar protein structure. The structure of the PMM2 and T237M are resembling, we can deduce that the change of aminoacid didn't have a great impact on the protein. However, it doesn't mean that there hasn't been any modifications between the interactions of the chains.

CROSS-SPECIES COMPARISON OF GENE AND PROTEIN SEQUENCES

The comparisons with different species, *Leishmania mexicana, a mamifer* and *Vicugna pacos* , a parasit gave us an idea of the conservation of the sequences so we could interpret its importance. Starting with Leishmania mexicana, we got a 70 percent identical sequence of nucleotides with the homo sapiens. But only 62 percent of the sequence could have been aligned. In this case, we know that more than the majority part of the sequence is very similar to homo sapiens. For Leishmania mexicana, it was possible only to align the 21

percent of the sequence, and from this sequence a 70 percent is identical to the Homo sapiens. Regarding the protein, 87 percent of the human sequence has been able to align with the Pacos. A 90 percent have the same positions of the aminoacids in both sequences, and a 92 (engloba) the identical or not but with similar chemical properties. So, it makes sense that protein positives have a higher percentage than the identity. On the other hand, we aligned 97 % of the sequence of Homo sapiens with Leishmania mexicana. And we obtained that 57 percent of the sequence of aminaocids are similar and a 71, which include also non-similar positions , have close chemical properties.

Overall,we obtained a bigger similitude of the human sequences with the Vicugna Pacos.

STRUCTURAL COMPARISON BETWEEN HUMAN AND L.MEXICANA

Another important aspect is that the three species in the position 11-17 of PMM2 had different nucleotides but in result they encoded for the same amino acid , called motif. This means that in this position that specific aminoacid has an importance for the protein, structurally or functionally, which means that the three different species conserved through evolution the same sequence of aminoacid (FDVDGTL). The fact that the sequences are in three different species means that it’s important to have them in order to maintain the function of the protein.

In the motif, there are Asp12 and Asp14, aminoacids that participate in the catalysis and magnesium binding. They were conserved in the three species so It suggests that these residues are functionally important. But again, conservation gives us evidence that the regions can be important functionally but we would need more direct experimentation to give a right conclusion.

LIMITATIONS

However, we have limitations that don't allow us to (be 100 sure) . We should take into account the genetic part of the gene.Bceasue, it could be more mutations of different genes that complement that leading to the disease or similar symptoms, also we should consider if it’s a(portador) . Also, regarding studies we only compared two species , giving us a reduced population to study.